Regret Metrics for Decision-Focused Learning

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2. **Optimize:** plug in and solve $z^*(f(X)) \in \arg\min_{z \in P} \langle f(X), z \rangle$.
3. **Evaluate:** the decision is evaluated at the *true* parameter, $\langle Y, z^*(f(X)) \rangle$.

Throughout, the downstream problem is **linear in** Y and $P \subseteq \mathbb{R}^d$ is the feasible set.

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A key observation

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Decision-Focused Learning (DFL) trains the model end-to-end against the downstream decision loss rather than a generic prediction loss. Techniques include:

- surrogate loss functions – SPO+ [Elmachtoub & Grigas, 2022], Fenchel–Young losses [Berthet et al., 2020];
- differentiation of a perturbed optimizer [Berthet et al., 2020];
- differentiation of a linear interpolation of the optimization mapping [Vlastelica et al., 2020].

A thorough survey is given in [Mandi et al., 2024].

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Any procedure to learn f that does not require the feasible region P is **Decision-Blind Learning (DBL)**.

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True Regret (against $f^*(X) = \mathbb{E}[Y \mid X]$)

$$\text{Regret}(f, f^*) = \mathbb{E}\left[\left\langle f^*(X), z^*(f(X)) \right\rangle\right] - \mathbb{E}\left[\left\langle f^*(X), z^*(f^*(X)) \right\rangle\right]$$

$\text{Regret}(f, f^*) =$ *0 for infinite compute, infinite data*.

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- On synthetic data, both are computable. On real data, only Look-Ahead Regret is.
- **Literature largely reports Look-Ahead Regret.**

The Regret Curves

A cartoon reconstruction of Elmachtoub & Grigas' shortest path problem (SPP) experimental results – de facto DFL benchmark:

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Figure 1.1: True Regret & Look-Ahead Regret for SPP

- Increased linear misspecification $\rightarrow$ harder problem
- Relative ranking of models across regrets is the same

Our Contributions

1. Look-Ahead Regret = True Regret + $\overbrace{\mathbb{E}\left[\langle f^*(X), z^*(f^*(X))\rangle - \langle Y, z^*(Y)\rangle\right]}^{\text{Look-Ahead Premium (LAP)}}$, where LAP is model-independent and benchmark-specific

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2. $\mathbb{E}\left[\text{LAP}\right]$ is convex, monotone and non-decreasing in the half-width of a bounded, mean-zero noise.
3. $\mathbb{E}\left[\text{LAP}\right]$ can behave non-monotonically w.r.t. linear misspecification in the data distribution.

Outline

True Regret vs. Look-Ahead Regret

Well-known result in literature:

Lemma 2.1 (The Perfect Regressor)

For a decision problem linear in the parameter Y, the ideal fitting function that minimizes expected decision loss is $f^*(X) = \mathbb{E}[Y \mid X]$

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Lemma 2.2 (Look-Ahead Regret Upper Bounds True Regret)

$\mathbb{E}[\langle Y, z^*(Y)\rangle] \leq \mathbb{E}\left[\left\langle f^*(X), z^*(f^*(X))\right\rangle\right]$, *and hence for* ***any*** f,

$$\text{Regret}(f, f^*) \leq \text{Regret}(f, Y).$$

Proof Outline: $\langle Y, z^*(Y)\rangle \leq \langle Y, z^*(f^*(X))\rangle$ by optimality, then apply expectation for claim 1. Rearrange and add $\mathbb{E}[f^*(X), z^*(f(X))]$ for claim 2.

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Expected Look-Ahead Premium, $\mathbb{E}\left[\text{LAP}\right]$

The advantage of an oracle that observes Y before making a decision, relative to the **best possible model** f^*.

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Expected Look-Ahead Premium, $\mathbb{E}\left[\text{LAP}\right]$

The advantage of an oracle that observes Y before making a decision, relative to the **best possible model** f^*.

Key property: $\mathbb{E}\left[\text{LAP}\right]$ depends only on the **problem setup**, not on f.

- Every model's Look-Ahead Regret curve is its True Regret curve **shifted upward by the same amount**.
- Therefore, differences in $\mathbb{E}\left[\text{LAP}\right]$ across instances can be mistaken for differences in **model performance** or **benchmark difficulty**.

Outline

The Shortest Path Benchmark [Elmachtoub & Grigas, 2022]

Find the shortest path on a 5×5 grid.

- $p = 5$ features per context, $d = 40$ edges in the response vector.
- Fixed random $B \in \mathbb{R}^{d \times p}$ with i.i.d. $\mathrm{Bernoulli}(0.5)$ entries.
- Contexts $x \sim N(0, I_p)$.

Cost generation (as in PyEPO [Tang & Khalil, 2024])

$$Y_j = \left[\frac{1}{3.5^{\mathsf{deg}}} \left(\frac{1}{\sqrt{p}} (Bx)_j + 3 \right)^{\mathsf{deg}} + 1 \right] \cdot \epsilon_j, \qquad \epsilon_j \sim \mathrm{Unif}[1-h,\, 1+h]$$

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- deg $\in \{1, 2, 4, 6, 8\}$ controls *linear misspecification*, with larger deg meant to be "harder."
- $h \in [0, 1]$ is the multiplicative noise half-width.

This design is now a standard benchmark: [Berthet et al., 2020], [Vlastelica et al., 2020], [Tang & Khalil, 2024], [Mandi et al., 2024], among others.

Our Shortest Path Experiment

Identical experiment to [Elmachtoub & Grigas, 2022] with SPO+ and Least Squares (LASSO) models. Training set size $N = 1000$. Noise half-width $h = 0.5$.

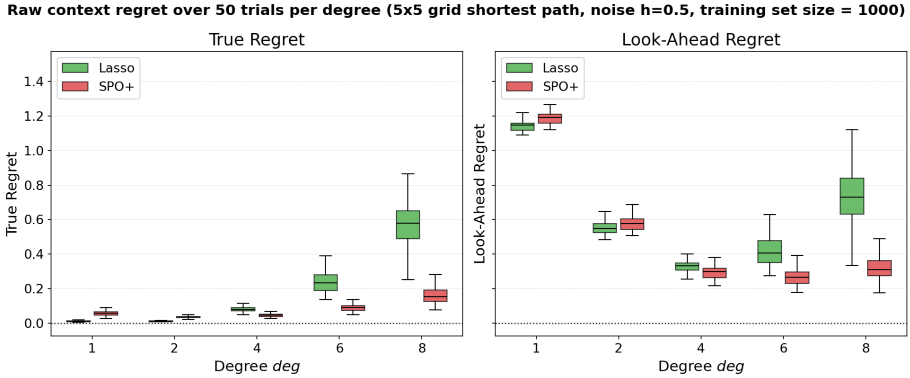

Figure 3.1: True Regret and Look-Ahead Regret for SPP

The original paper reports normalized versions of exactly our two metrics: "noiseless regret" $\propto \text{Regret}(f, f^*)$ and "noisy regret" $\propto \text{Regret}(f, Y)$.

LAP Distorts Interpretation of Regret

Look-Ahead Regret doesn't just upper bound True Regret:

Regret(f, f^*) — as expected

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This distortion is because of the **Look-Ahead Premium,** since:

$$\text{Regret}(f, Y) = \text{Regret}(f, f^*) + \mathbb{E}\,[\text{LAP}]$$

Expanding LAP

Write

$$\epsilon = \mathbf{1} + h\gamma, \qquad \gamma \sim \mathrm{Unif}[-1,1]^d,$$

so that

$$Y = \underbrace{f^*(X)}_{\text{mean}} + \underbrace{hf^*(X) \odot \gamma}_{\text{noise}}.$$

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For a fixed realization of (X, γ),

$$\begin{aligned} \mathsf{LAP} &= \left\langle Y,\, z^*(f^*(X)) - z^*(Y) \right\rangle \\ &= \max_{z \in P} \left\langle f^*(X) + hf^*(X) \odot \gamma,\, z^*(f^*(X)) - z \right\rangle. \end{aligned}$$

LAP vs. the noise half-width h

Theorem 3.1

Fix a degree deg*. For every realization of* (X, γ), $\mathsf{LAP}(h)$ *is convex and non-decreasing on* $h \in [0, 1]$*. Consequently* $\mathbb{E}\left[\mathsf{LAP}(h)\right]$ *is convex and non-decreasing on* $[0, 1]$*.*

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Since both hold for any draw of (X, γ), they hold in expectation.

LAP vs. the polynomial degree deg

Observe:

$$Y = f^*(X) \odot (1 + h\gamma) = \underbrace{f^*(X)}_{\text{mean}} + \underbrace{hf^*(X) \odot \gamma}_{\text{noise, mean } 0 \text{ given } X}$$

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Setting $d_1 = d_2$ recovers the original benchmark.

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h argument does not extend to deg. *We leave this analysis for future work!*

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Experimental Protocol

For each problem instance we isolate LAP from any model entirely:

- Draw N fresh test contexts X and responses Y.
- Solve the decision problem **exactly** under both $f^*(X)$ and Y.
- Average $\left\langle f^*(X), z^*(f^*(X)) \right\rangle - \left\langle Y, z^*(Y) \right\rangle$ over the test set.

No training is involved — the quantity is model-free.

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Three classical DFL benchmarks:

1. **Shortest Path** [Elmachtoub & Grigas, 2022] — de facto DFL benchmark
2. **Traveling Salesman** [Tang & Khalil, 2024] — similar polynomial to SPP, part of PyEPO benchmark suite.
3. **Portfolio Optimization** [Elmachtoub & Grigas, 2022, Sec. 6.2] — continuous, convex QCQP + other experiment in SPO+ paper.

Shortest Path Problem (SPP)

Setup as in Section 3: $p = 5$, $d = 40$, fixed B with i.i.d. $\mathrm{Bernoulli}(0.5)$ entries. Sweep deg $\in \{1, 2, 4, 6, 8\}$ and $h \in \{0.01, 0.25, 0.5, 0.75, 0.99\}$ with $N = 1000$ test contexts per pair.

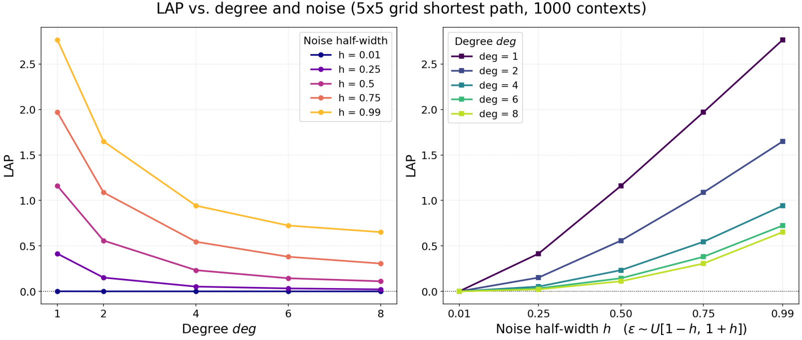

Figure 4.1: Estimated $\mathbb{E}[\mathrm{LAP}]$ as a function of deg and of h.

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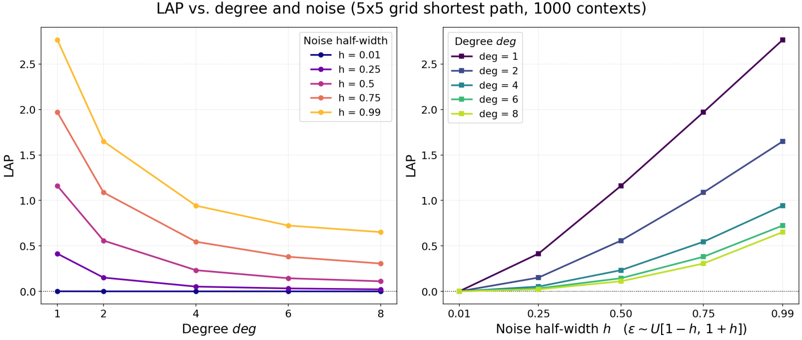

Figure 4.1: Estimated $\mathbb{E}\,[\text{LAP}]$ as a function of deg and of h.

Monotone, convex and non-decreasing in h, exactly as Theorem 3.1 predicts.
Monotone, convex and non-increasing in deg.

SPP: Degree Disassociation

We empirically estimate the effect of Y_{d_1,d_2} as the true response. Fix $h = 0.5$ and estimate decision flip rate %age (DFR) and $\mathbb{E}[\text{LAP}]$ for every (d_1, d_2) pair.

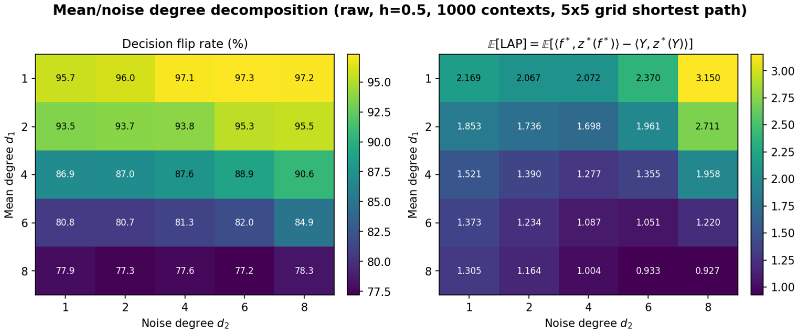

Figure 4.2: DFR and estimated $\mathbb{E}[\text{LAP}]$ as a function of (d_1, d_2).

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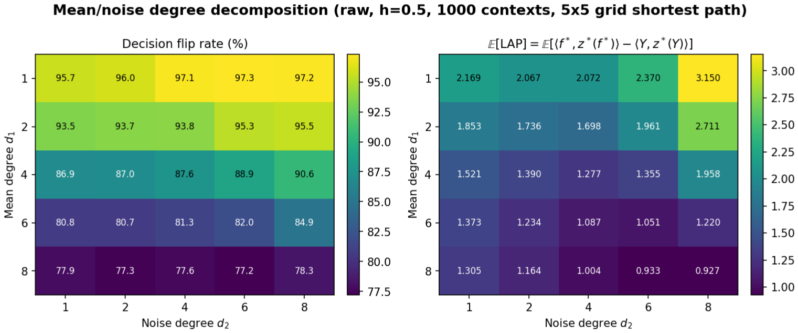


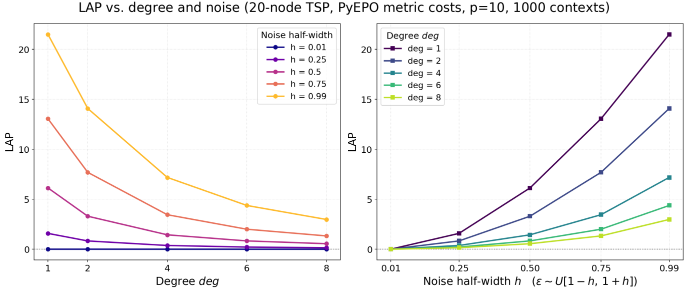
Figure 4.2: DFR and estimated $\mathbb{E}\left[\text{LAP}\right]$ as a function of (d_1, d_2).

Fixed d_2: DFR and $\mathbb{E}\left[\text{LAP}\right]$ decrease.
Fixed d_1: DFR increases for small d_1, decreases then increases for large d_1.
$\mathbb{E}\left[\text{LAP}\right]$ decreases, then increases.

Traveling Salesman Problem (TSP)

Symmetric TSP on v nodes. Following [Tang & Khalil, 2024], edge j's cost adds a **noise-free** Euclidean term to a noisy polynomial term:

$$Y_j = \mathrm{eucl}_j + \left[\frac{1}{3^{\mathsf{deg}-1}} \left(\frac{1}{\sqrt{p}} (Bx)_j + 3 \right)^{\mathsf{deg}} \right] \cdot \epsilon_j, \qquad \epsilon_j \sim \mathrm{Unif}[1-h, 1+h]$$



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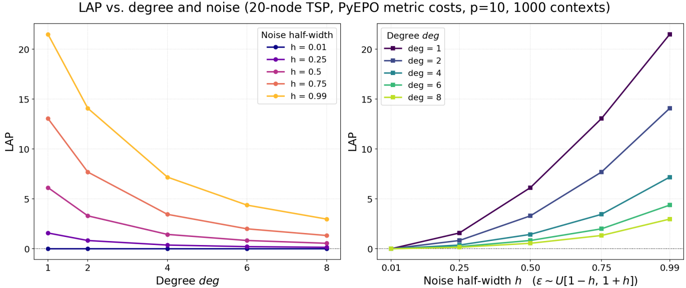

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Monotone, convex and non-decreasing in h, exactly as Theorem 3.1 predicts.
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Portfolio Optimization (PO): Setup

Mean–variance (Markowitz) problem on $d = 50$ assets [Elmachtoub & Grigas, 2022, Sec. 6.2]. A decision z allocates weights subject to budget and risk constraints:

$$P := \left\{z \in \mathbb{R}^d : \ z^\top \Sigma z \le \gamma, \ \mathbf{1}^\top z \le 1, \ z \ge 0\right\}$$

Writing $-Y$ as the excess returns, we solve the minimization $\min_{z \in P} \langle Y, z \rangle$.

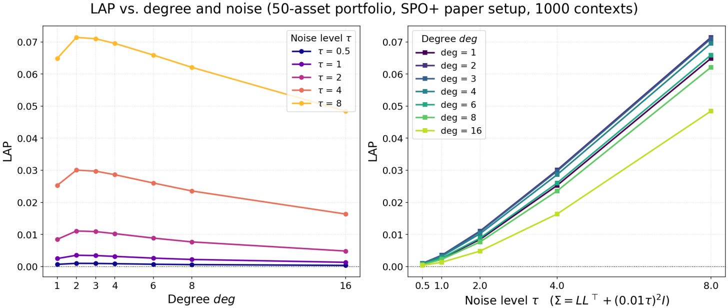
Factor model

$$-f^*(X) = \left(\frac{0.05}{\sqrt{p}}(B^* x)_j + 0.1^{1/\mathsf{deg}}\right)^{\mathsf{deg}}, \qquad Y = f^*(X) - L\phi - 0.01\tau\epsilon$$

with $\phi \sim N(0, I_4)$, $\epsilon \sim N(0, I_d)$, and $\Sigma = LL^\top + (0.01\tau)^2 I_d$.

P is **continuous and convex** — $z^*(\cdot)$ is solved as a QCQP with a conic solver, not combinatorially. deg $\in \{1, 2, 3, 4, 6, 8, 16\}$, $\tau \in \{0.5, 1, 2, 4, 8\}$, $N = 1000$.

Portfolio Optimization: Results



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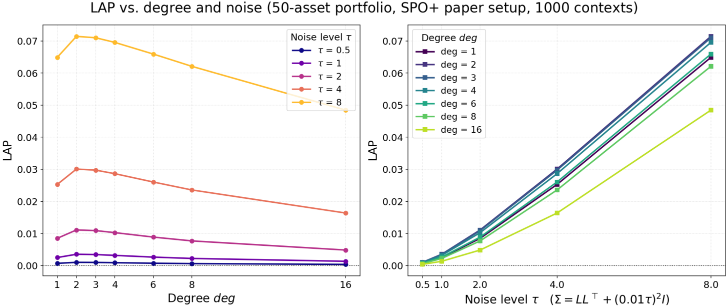

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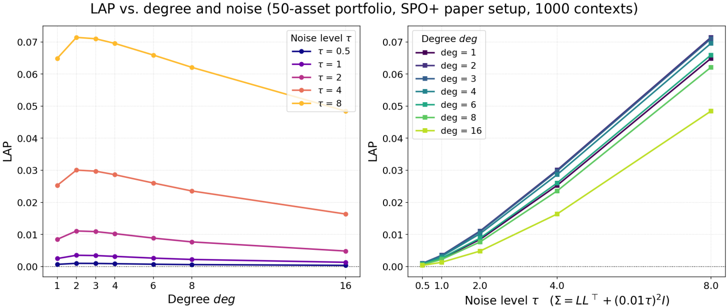

Figure 4.4: Estimated $\mathbb{E}\left[\text{LAP}\right]$ as a function of deg and noise level τ.

deg**: Parabolic!** Unexpected result compared to our prior findings, but note the difference in cost generation.

τ**: Monotone, convex and non-decreasing,** exactly as Theorem 3.1 predicts. Holds because $Y = f^*(X) - L\phi - 0.01\tau\epsilon$ is still an affine function in τ.

Outline

Takeaways

1. Look-Ahead Regret $\text{Regret}(f, Y)$ **always upper bounds** True Regret $\text{Regret}(f, f^*)$, and the gap $\mathbb{E}[\text{LAP}]$ is completely independent of the model.

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Wherever possible, report both regrets + acknowledge when the metric of interest is affected by the choice of regret.

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Questions?

Backup slide 1: Notation

- (X, Y): the random context–response pair; (x, y) a realization.
- f: a fitting function, producing prediction $f(X)$.
- $f^*(X) := \mathbb{E}[Y \mid X]$: the **true conditional mean**, or *perfect regressor*.
- $P \subseteq \mathbb{R}^d$: the feasible set of the downstream problem.
- $z^*(y) \in \arg\min_{z \in P} \langle y, z \rangle$: the induced decision.

Decision loss

The decision loss of f is the cost actually incurred by acting on $f(X)$:

$$\mathbb{E}\left[\left\langle Y,\ z^*(f(X))\right\rangle\right].$$

This is evaluated under the *true* parameter Y, not under the prediction.

Backup slide 2: The Perfect Regressor

Lemma 5.1 (The Perfect Regressor)

For a decision problem linear in the parameter Y, the ideal fitting function that minimizes expected decision loss is $f^*(X) = \mathbb{E}[Y \mid X]$.

Proof: Since $z^*(f(X))$ is a deterministic function of X, the tower rule gives

$$\mathbb{E}\left[\left\langle Y, z^*(f(X)) \right\rangle\right] = \mathbb{E}\left[\mathbb{E}\left[\left\langle Y, z^*(f(X)) \right\rangle \,\middle|\, X\right]\right] = \mathbb{E}\left[\left\langle \mathbb{E}[Y \mid X],\ z^*(f(X)) \right\rangle\right].$$

Fix a realization $X = x$. By optimality of $z^*(\mathbb{E}[Y \mid X = x])$ under the cost vector $\mathbb{E}[Y \mid X = x]$,

$$\left\langle \mathbb{E}[Y \mid X = x], z^*(f(x)) \right\rangle \ \geq\ \left\langle \mathbb{E}[Y \mid X = x], z^*(\mathbb{E}[Y \mid X = x]) \right\rangle,$$

with equality at $f(x) = f^*(x)$. This holds for every x. □

So a fitting function f should be compared against f^* and not Y.

Backup slide 3: Look-Ahead Regret Always Upper Bounds Regret

Lemma 5.2

$\mathbb{E}\left[\langle Y, z^*(Y)\rangle\right] \;\le\; \mathbb{E}\left[\big\langle f^*(X), z^*(f^*(X))\,\big\rangle\right]$, *and hence for* ***any*** f,

$$\mathrm{Regret}(f, f^*) \;\le\; \mathrm{Regret}(f, Y).$$

Proof: By optimality, $\langle Y, z^*(Y)\rangle \le \langle Y, z^*(f^*(X))\rangle$ almost surely. Taking expectations and conditioning on X,

$$\mathbb{E}\left[\langle Y, z^*(Y)\rangle\right] \le \mathbb{E}\left[\big\langle \mathbb{E}\left[Y \mid X\right], z^*(f^*(X))\,\big\rangle\right] = \mathbb{E}\left[\big\langle f^*(X), z^*(f^*(X))\,\big\rangle\right].$$

Adding $\mathbb{E}\left[\langle f^*(X), z^*(f(X))\rangle\right]$ to both sides and using $\mathbb{E}\left[\langle f^*(X), z^*(f(X))\rangle\right] = \mathbb{E}\left[\langle Y, z^*(f(X))\rangle\right]$ (again by the tower rule) gives the second claim. □

Backup slide 4: A Lower-Variance Estimator

Remark 5.1 (Estimating Look-Ahead Regret)

When f^* is known, Regret(f, Y) *may be estimated by either*

$$\mathbb{E}\left[\langle Y, z^*(f(X))\rangle - \langle Y, z^*(Y)\rangle\right] \quad \textit{or} \quad \mathbb{E}\left[\langle f^*(X), z^*(f(X))\rangle - \langle f^*(X), z^*(Y)\rangle\right].$$

The two agree in expectation, but the latter has **lower variance** *as an empirical estimator: it removes the randomness carried by the noisy realization Y.*

This is practically useful — it sharpens the boxplots in the experiments later without changing the quantity being measured.

Backup slide 5: Varying the Mean and the Noise Separately

Varying the mean (d_2 fixed). With $C := hf^*(X; d_2) \odot \gamma$, $\mathbb{E}[C] = 0$:

$$\mathbb{E}[\mathsf{LAP}] = \mathbb{E}\left[\left\langle f^*(X; d_1), z^*(f^*(X; d_1)) \right\rangle\right] - \mathbb{E}\left[\left\langle f^*(X; d_1), z^*(f^*(X; d_1) + C) \right\rangle\right]$$

An **additive** noise regime with fixed perturbation C. As d_1 grows, the spread of path costs widens and the top-ranked path becomes harder to overtake — pushing LAP *down*.

Varying the noise (d_1 fixed). With $C := f^*(X; d_1)$:

$$\mathbb{E}[\mathsf{LAP}] = \mathbb{E}[\langle C, z^*(C) \rangle] - \mathbb{E}\left[\left\langle C,\ z^*(C + hf^*(X; d_2) \odot \gamma) \right\rangle\right]$$

Only the perturbation magnitude moves; larger d_2 inflates it — pushing LAP *up*.

The observed parabola in deg **is these two forces competing.** (In progress.)

Backup slide 6: Ongoing and Future Work

- **Strict monotonicity in h.** Conditions guaranteeing $\mathsf{LAP}(h) > 0$ for $h > 0$ to show that positive noise can flip the argmin with positive probability.
- **Explain $\mathbb{E}[\mathsf{LAP}]$ against deg for the SPP.** Rigorously prove the empirical finding that $\mathbb{E}[\mathsf{LAP}]$ decreases as deg increases for any noise h.
- **Formalize the strict concavity of $\mathbb{E}[\mathsf{LAP}]$ against deg for PO.** A proposition characterizing exactly when behavior of $\left(\frac{0.05}{\sqrt{p}}(B^*x) + 0.1^{1/\mathsf{deg}}\right)^{\mathsf{deg}}$ breaks monotonicity.
- **The mean/noise decomposition of deg.** Complete the analysis of Y_{d_1,d_2} to quantify how much of the parabola each effect contributes.
- **Real data.** When f^* is unavailable, can $\mathbb{E}[\mathsf{LAP}]$ be *estimated* and subtracted off, restoring a usable proxy for $\mathsf{Regret}(f, f^*)$?

Backup slide 7: About Me

- **Me:** Shammas Ahmed, originally from Lahore, Pakistan
- **Education:** Undergraduate Senior at Rice University (Houston, TX). Completing a B.S. in Operations Research
- **Interests:** Combinatorial optimization and graph problems, online optimization under competition
- **Hobbies:** Playing squash, listening to DJ sets, exploring matcha places in LA
- **Fun fact:** This is my first time in California!